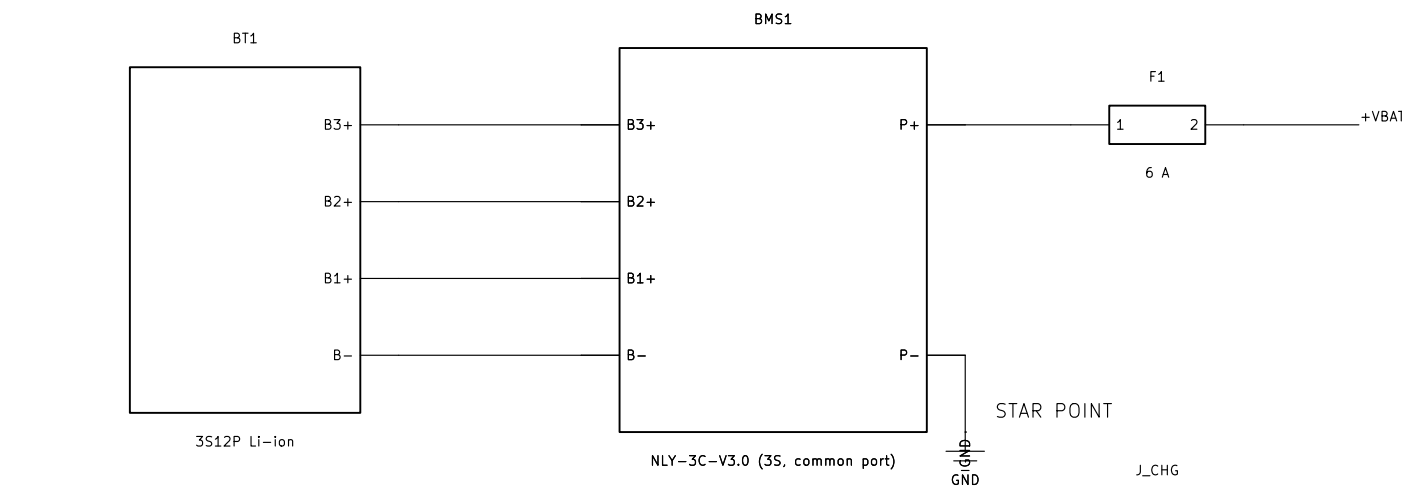


ZONE A POWER DISTRIBUTION

Rev 3.0: the 3S BMS enters here



CONNECT THE TAPS IN THIS ORDER: B- then B1+ then B2+ then B3+ Out of order, the protection IC sees most of the pack across one stage and dies. Solder them to the pack FIRST, meter the connector, and only then plug it into the board.

B1+ and B2+ carry milliamps -- 22-24 AWG.
B3+ and B- carry the FULL pack current -- run those two thick.

*** THE STAR POINT MOVED IN REV 3.0 ***

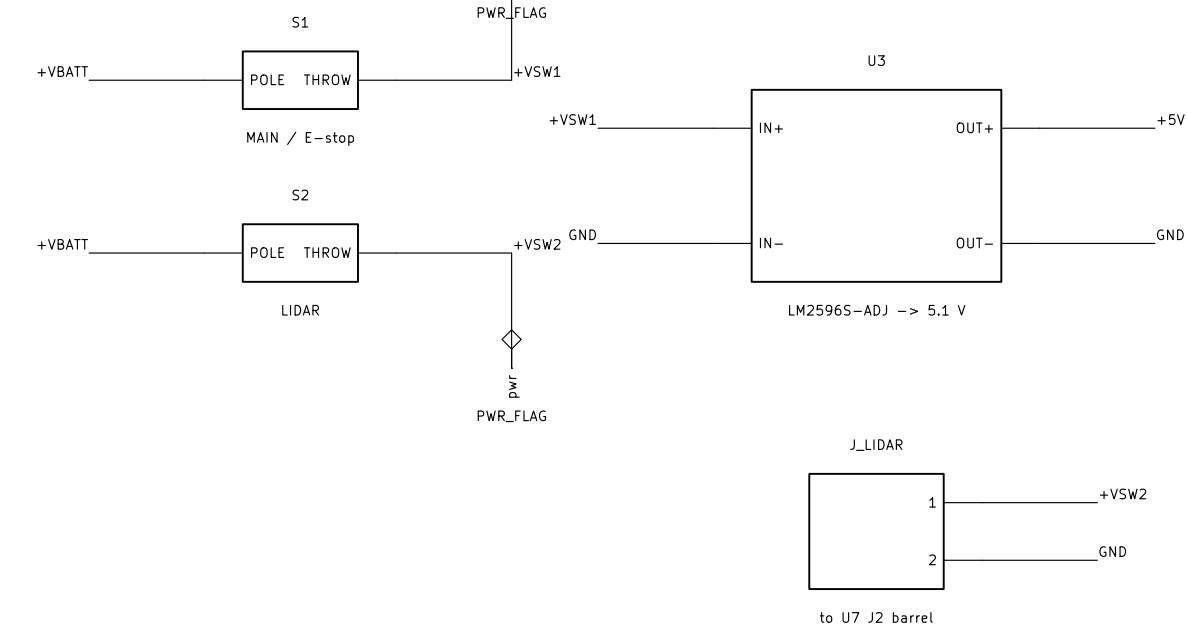
Every ground in this rig lands on BMS P-, never on the pack's B-. A return wired to B- bypasses the FETs: that load is unprotected, AND it keeps draining the pack after the BMS has cut off. EXACTLY ONE wire touches B-, and it is the one from the pack.

STAR POINT

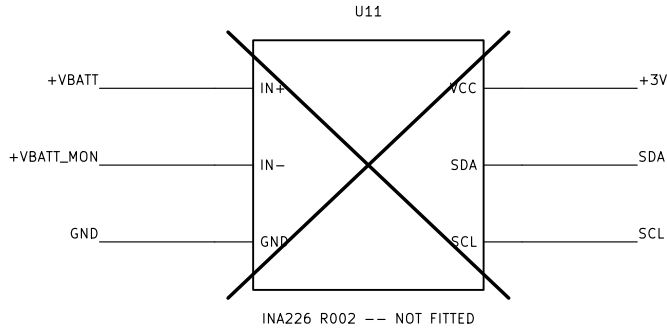
GND

Charger: 12.6 V CC-CV ONLY.
13.8-14.4 V open circuit is a lead-acid charger and must not touch this pack. Measure it before it is ever plugged in.

S1 and S2 tap the same fused node in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.



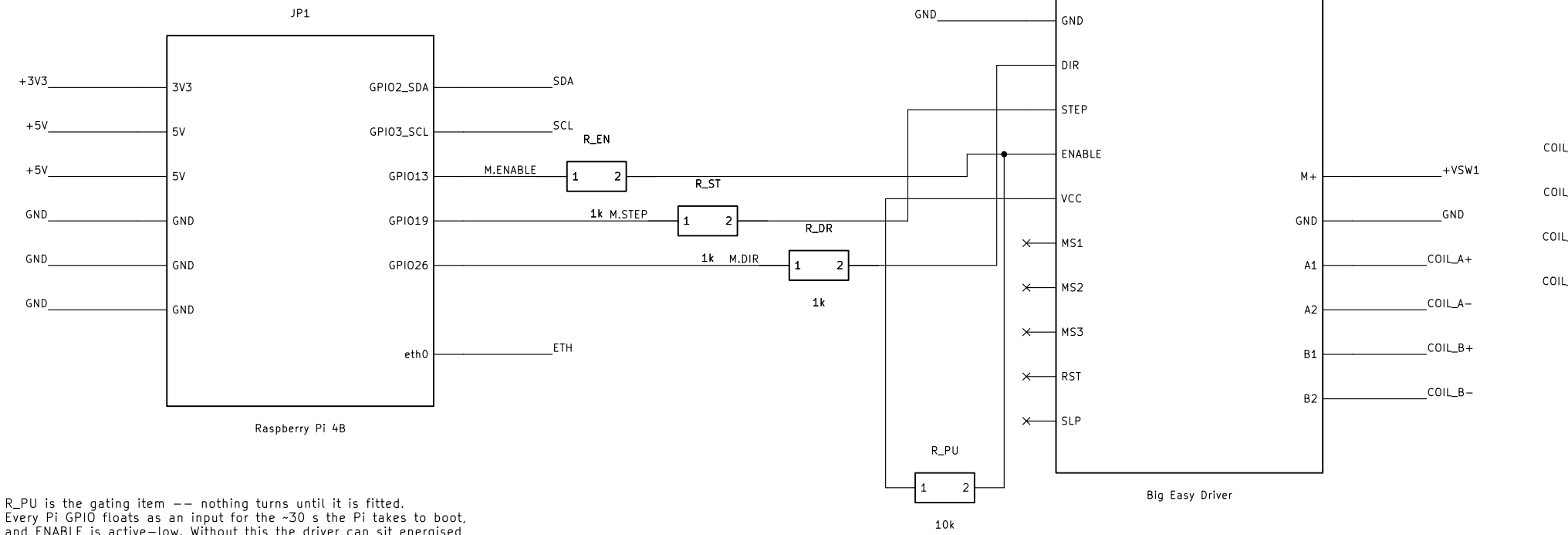
U6 IS DELETED IN REV 3.0.
A buck needs -1.5 V of headroom, so an LM2596 set to 12 V on a 12 V pack never regulates -- it just sags under load, which is exactly when the motor needs it. M+ now takes +VSW1 directly; the Big Easy Driver accepts 8-35 V.



INA226 R002 -- NOT FITTED
DNP -- ordered, never connected. Decide its position NOW: the shunt sits in series with the pack lead, so retrofiting means cutting the harness you are about to make. R002 (0.002 ohm), NOT R100 -- R100 is good for 0.8 A and this rig pulls -3 A. 3V3 only, never 5 V, or the module's I2C pull-ups put 5 V on GPIO2/GPIO3.

ZONE B MOTOR CHAIN

unchanged from Rev 2.0 except M+

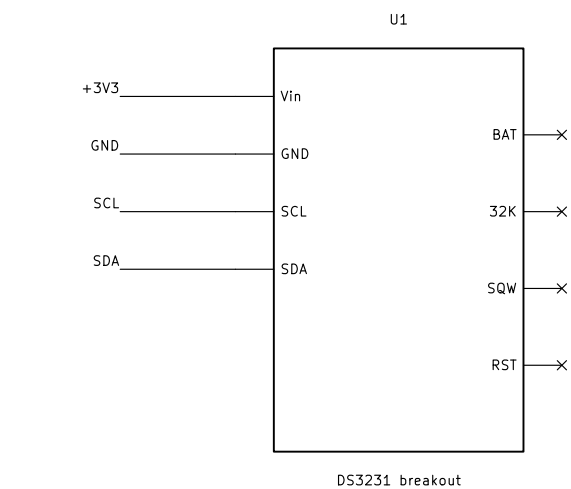


R_PU is the gating item -- nothing turns until it is fitted. Every Pi GPIO floats as an input for the ~30 s the Pi takes to boot, and ENABLE is active-low. Without this the driver can sit energised through the whole of boot with nothing in control of it. It goes on the DRIVER side of R_EN, and pulls up to the driver's own VCC -- not the Pi's 3V3 -- so it still holds with the Pi unplugged. Set the APWR jumper to 3.3 V and MEASURE VCC before wiring to it.

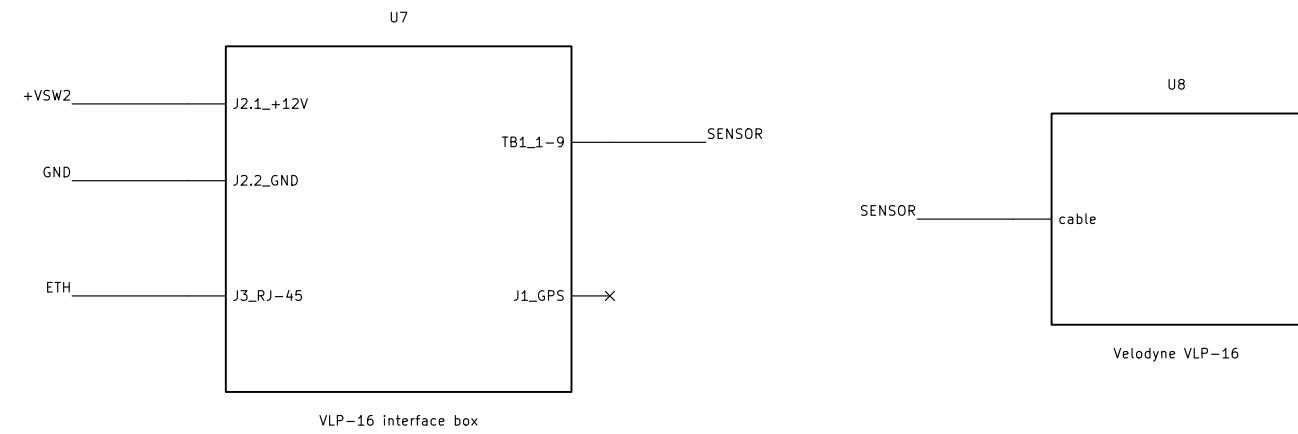
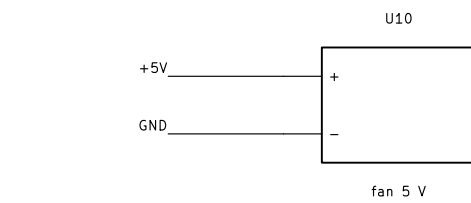
A1/A2 must be the two ends of ONE coil. B1/B2 the two ends of the other. Cross them and the motor buzzes, heats and does not turn. Which way round a pair goes only reverses direction, and direction is a software setting (TSLPIE_DIR_FORWARD) -- do not chase it with a soldering iron. Find the pairs with a meter on continuity. motor disconnected: one coil reads a few ohms, the other open.

ZONE C CAPTURE, CLOCK, COOLING

unchanged from Rev 2.0



Vin from header pin 1 (3V3). NEVER pin 2 or 4. This board's I2C pull-ups reference Vin, so 5 V on Vin puts 5 V on GPIO2 and GPIO3, which are not 5 V tolerant. Check it before you trust it: power the board with SDA and SCL not yet fitted and measure SDA to GND -- it must read about 3.3 V. BAT / 32K / SQW / RST: no connection, on purpose.



eth0 stays on 192.168.1.100 and the phone hotspot must not. The sensor never touches the GPIO header -- it is Ethernet the whole way. TB1 inside U7 takes the VLP-16 factory cable: 1 Gnd/Shield, 2 +12 Vdc, 3 GPS_Pulse_CNT, 4 GPS_RX_CNT, 5 Eth TX+, 6 Eth TX-, 7 Eth RX+, 8 Eth RX-, 9 GPS return.

ACCEPTANCE TEST -- the same measurement that diagnosed the 4S fault

1. B- to P- must read a few MILLivolts, ~0.55 V means both FETs are off and current is sneaking through the body diodes. That is the exact 4S symptom, and it is what strangled a healthy pack.
2. P+ to P- must equal B3+ to B- within a few mV.
3. Each group -- B- to B1+, B1+ to B2+, B2+ to B3+ -- within 50 mV of the others. 12.22 V total is 4.07 V/cell: a full, healthy pack.
4. Only then close S1, and only with the motor uncoupled.

TLS Pie -- Rev 3.0 -- generated by kicad/make_kicad_schematic.py Edit the script, not this file. Supersedes Rev 2.0 (WIRING_REV2.html). Source of truth for GPIO assignment is tls_stepper.py.

GPIO assignment is owned by tls_stepper.py
Generated by kicad/make_kicad_schematic.py -- edit the script
Adds the 3S BMS. Star point moves to BMS P-. U6 deleted.
Supersedes Rev 2.0 of 2026-08-09

Kizim Robotics
Sheet: /
File: TLS_Pie.kicad_sch
Title: TLS Pie -- Rev 3.0 wiring schematic

Size: A2 Date: 2026-08-11 Rev: 3.0
KiCad E.D.A. 10.0.5 Id: 1/1